SATYANARAYANA REDDY T

SOFTWARE RF ENGINEER · QT/QML · C#/.NET · WINDOWS / LINUX (UBUNTU)

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PROFESSIONAL SUMMARY

Software RF Engineer with 9+ years of overall experience — 6+ years in technical support and 3.5+ years in application coding. Skilled at understanding client requirements for RF Test Jigs and developing applications over RS-232, RS-485, and TCP/IP, with FTP-based file transfer and automated report generation using Qt ActiveX controls, C# ClosedXML, and CsvHelper. Currently builds defence and space software at Astra Microwave — ELINT/EW data-visualisation GUIs (ScottPlot 5, PostgreSQL), MIL-STD-1553 DDC applications, and RF automated test equipment using Keysight and NI instruments over SCPI. Combines strong customer-support troubleshooting with GUI architecture (MVC/MVVM, custom QML components, multithreading) and owns work from requirements through coding, review, testing, documentation, and release.

CORE SKILLS

Languages: C++, C#, Qt/QML, SQL, JavaScript, Python (Flask, scripting), HTML
Frameworks & Libraries: Qt Widgets, Qt Quick/QML, .NET 8.0 (WPF, WinForms, MVVM), ScottPlot 5, OpenCV
Platforms: Windows, Ubuntu Linux, Android, Raspberry Pi, Jetson Nano
Databases: PostgreSQL, SQLite
Development Tools: Visual Studio 2019/2022/2026, Qt 5.15/6.0/6.3, VS Code, Git/GitHub, Doxygen
Protocols & Interfaces: TCP/IP, UDP, RS-232, RS-485, FTP, UART, MIL-STD-1553 (DDC), SCPI
RF Test & Reporting: RF test jigs, Keysight/NI ATE, SCPI control, Qt ActiveX, ClosedXML, CsvHelper
Modernization & Cloud: Docker, Kubernetes (basics), Kafka/CDC, AWS EC2/S3/WorkSpaces, VMware
ITSM & Support: Jira, ServiceNow, SMAX, HP Service Manager, Siebel, Citrix, Salesforce, Active Directory, Bash scripting

PROFESSIONAL EXPERIENCE

Software RF Engineer — Qt/QML & C#

Nov 27, 2023 – Present

Astra Microwave Products Ltd — Hyderabad, India

- Understand client requirements and develop applications for RF Test Jigs — device links over RS-232, RS-485, and TCP/IP, FTP file transfer, and automated report generation using Qt ActiveX controls and C# (ClosedXML, CsvHelper).
- Build defence and space software — ELINT/EW data visualisation (ScottPlot 5, PostgreSQL), MIL-STD-1553 DDC interfaces, and RF automated test equipment controlled over SCPI.
- Develop GUI and desktop applications in C#/.NET 8.0 (WPF, WinForms, MVVM) and Qt/QML for Windows and Ubuntu Linux.
- Analyse requirements with cross-functional teams, implement features and defect fixes, and support releases; use Git for source control and participate in code reviews.
- Perform application testing and cross-platform troubleshooting on Windows and Linux builds; document modules and contribute unit/component tests.

Qt/QML GUI Development Engineer

Jun 2020 – Nov 2023

Epigon Media Technologies Pvt Ltd — Bangalore, India

- Designed and developed cross-platform Qt/QML applications for Windows, Ubuntu Linux, and Android with consistent UI behaviour.
- Built reusable QML components and Qt Widgets using MVC/MVVM patterns, custom models, and multithreaded design for responsive live-data interfaces.
- Integrated devices and services over TCP/UDP sockets, UART, REST APIs, and SQLite; used OpenCV and FFmpeg/GStreamer for live video, image capture, and processing.
- Owned features end to end — requirement analysis, technical design, coding, code review, functional testing, Doxygen documentation, and release support.
- Developed Python Flask REST services and a Qt client for IBM Watson chatbot-based real-time monitoring.

Technical Support Engineer

Sep 2017 – May 2020

Epigon Media Technologies Pvt Ltd — Bangalore, India

- Provided Tier 1/Tier 2 support to Ubuntu Linux and Windows customers via phone, email, chat, and ITSM tools (Jira, ServiceNow, SMAX, HP Service Manager, Siebel), meeting SLA response and resolution targets.
- Diagnosed installation, configuration, hardware, software, and network issues (TCP/IP, DNS, DHCP, VPN) using ping, telnet, tracert, and Wireshark.
- Administered Ubuntu/Linux servers — package and patch management, user accounts and permissions, backups, system monitoring, and Apache/Tomcat deployments.
- Supported Active Directory administration, AWS WorkSpaces, and S3 file retrieval; documented fixes and contributed knowledge-base articles.
- Escalated complex cases to Tier 3 with clear reproduction steps; collaborated with US and Canada teams on Project Fi deployment and used Citrix for remote support.

KEY PROJECTS

ELINT & Electronic Warfare GUI Software
Nov 2023 – Present

C#/.NET 8.0, WPF, WinForms, MVVM, ScottPlot 5, PostgreSQL | Windows

- Develop desktop GUI modules for a satellite-based Electronic Intelligence (ELINT) and Electronic Warfare system covering direction-of-arrival characterisation and signal monitoring.
- Build real-time data-visualisation screens with ScottPlot 5 — spectrum, waterfall, and trend plots — with PostgreSQL storing configuration, measurement, and analysis data.
- 3.5+ years of hands-on C# coding across UI, data access, plotting, and instrument-control layers.

MIL-STD-1553 DDC Applications (Space & Defence)
Nov 2023 – Present

C#/.NET 8.0, DDC MIL-STD-1553 interface cards, WPF | Windows

- Develop applications using Data Device Corporation (DDC) MIL-STD-1553 interfaces for space and defence programmes, including bus configuration, message scheduling, and data capture.
- Implement message encode/decode, error handling, and logging for 1553 traffic; support hardware integration and system testing with engineering teams.

RF System Automated Test Equipment (ATE)
Nov 2023 – Present

C#/.NET 8.0, SCPI, Keysight & NI instruments, WPF | Windows

- Develop RF automated test equipment software that configures and controls Keysight and National Instruments (NI) instruments over SCPI commands.
- Automate measurement sequences, instrument set-up, result capture, and pass/fail reporting.

RF Test Jig Client Applications
Nov 2023 – Present

C#/.NET 8.0, Qt, RS-232, RS-485, TCP/IP, FTP, ClosedXML, CsvHelper | Windows

- Understood client requirements and developed applications for RF Test Jigs with jig and instrument control over RS-232, RS-485, and TCP/IP.
- FTP file transfer between jig and application; report generation via Qt ActiveX, ClosedXML (Excel), and CsvHelper (CSV).

Eye Tracking & Image Processing Software (ETIM)
Sep 2022 – Nov 2023

Qt, QML, OpenCV | Raspberry Pi, Windows, Ubuntu Linux

- Developed a Qt/QML application for live camera preview, autofocus, auto-capture, noise removal, masking, and image fine-tuning using OpenCV image-processing algorithms.
- Tuned capture and processing parameters to deliver consistent image quality on Raspberry Pi and Windows; built operator screens for reviewing, capturing, and exporting processed images.

Power Quality Measurement (PQM)
Oct 2021 – Sep 2022

Qt, Python, Flask, SQLite, HTML | Jetson Nano, BF518, Windows, Ubuntu Linux, Android

- Built a cross-platform Qt GUI displaying real-time three-phase voltage, current, active/reactive power, power factor, frequency, and energy data, with IEC 61000-4-30 Class S half-cycle and 10/12-cycle RMS, dip/swell, VTHD, and ITHD measurements.
- Developed a responsive HTML dashboard and a Python Flask REST API connecting the application to an IBM Watson chatbot for real-time monitoring.

Content Management & Distribution System (CMDS)
Jan 2018 – Sep 2021

C++, Qt 5.12 | Ubuntu Linux

- Built Qt client workflows and REST API integration for Broadcast Operation Center (BOC) and Edge servers, including channel creation and bandwidth configuration.
- Implemented immediate and scheduled file/live-video transmission workflows and edge-server registration, lookup, and monitoring across distributed locations.

EDUCATION

Bachelor of Technology (B.Tech), Electrical & Electronics Engineering — 2012

Madhira Institute of Technology & Science, Jawaharlal Nehru Technological University, Hyderabad | 70%

Intermediate (MPC) — Sahithi & Vanitha Junior College, Board of Intermediate, Andhra Pradesh — 2008 | 71%

SSC — Bharath English Medium School, Andhra Pradesh — 2006 | 69%

ADDITIONAL INFORMATION

Languages: English, Telugu

Availability: Open to rotational shifts, on-call support, and travel

Strengths: Root-cause troubleshooting · Customer communication · Ownership · Quick learner